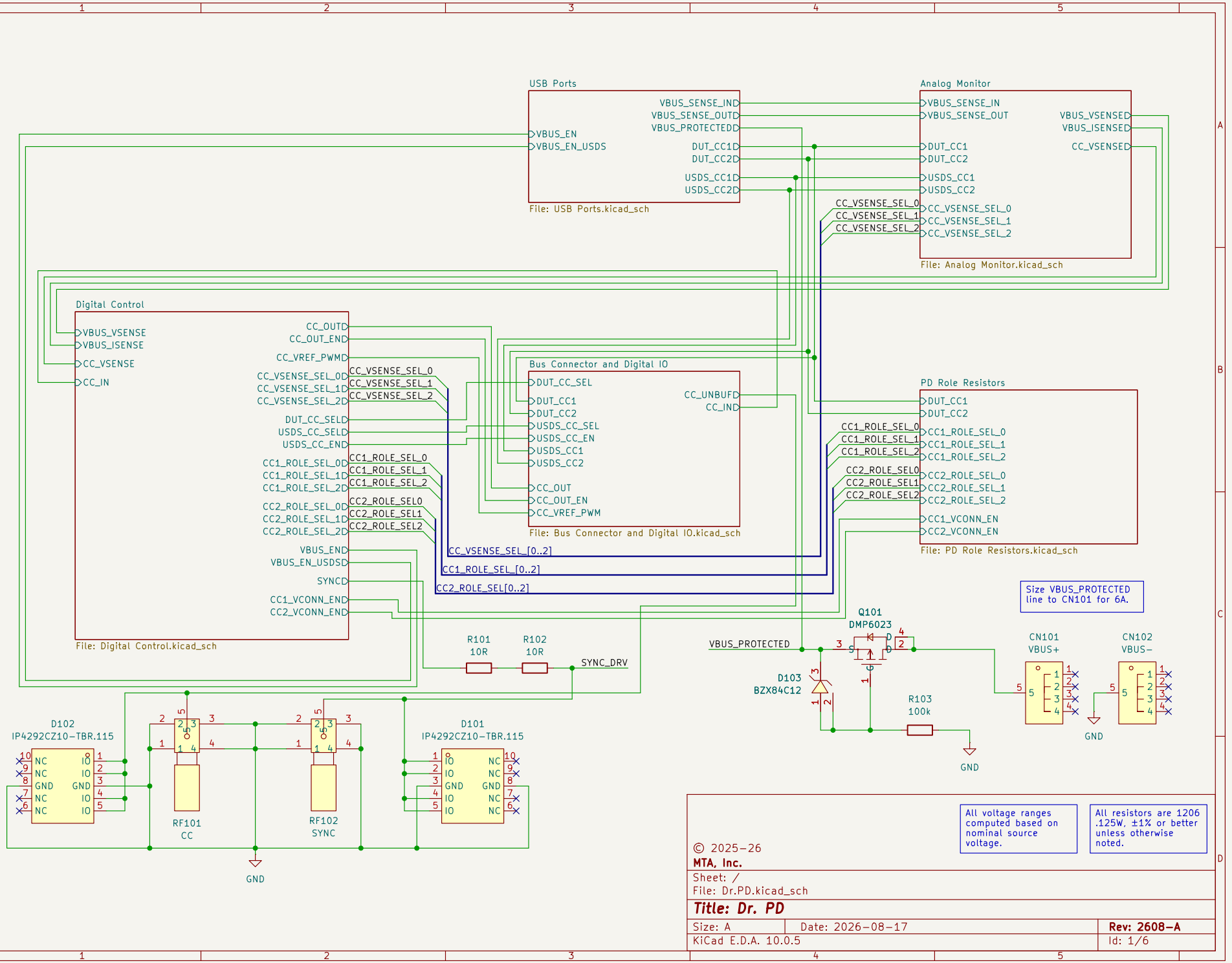
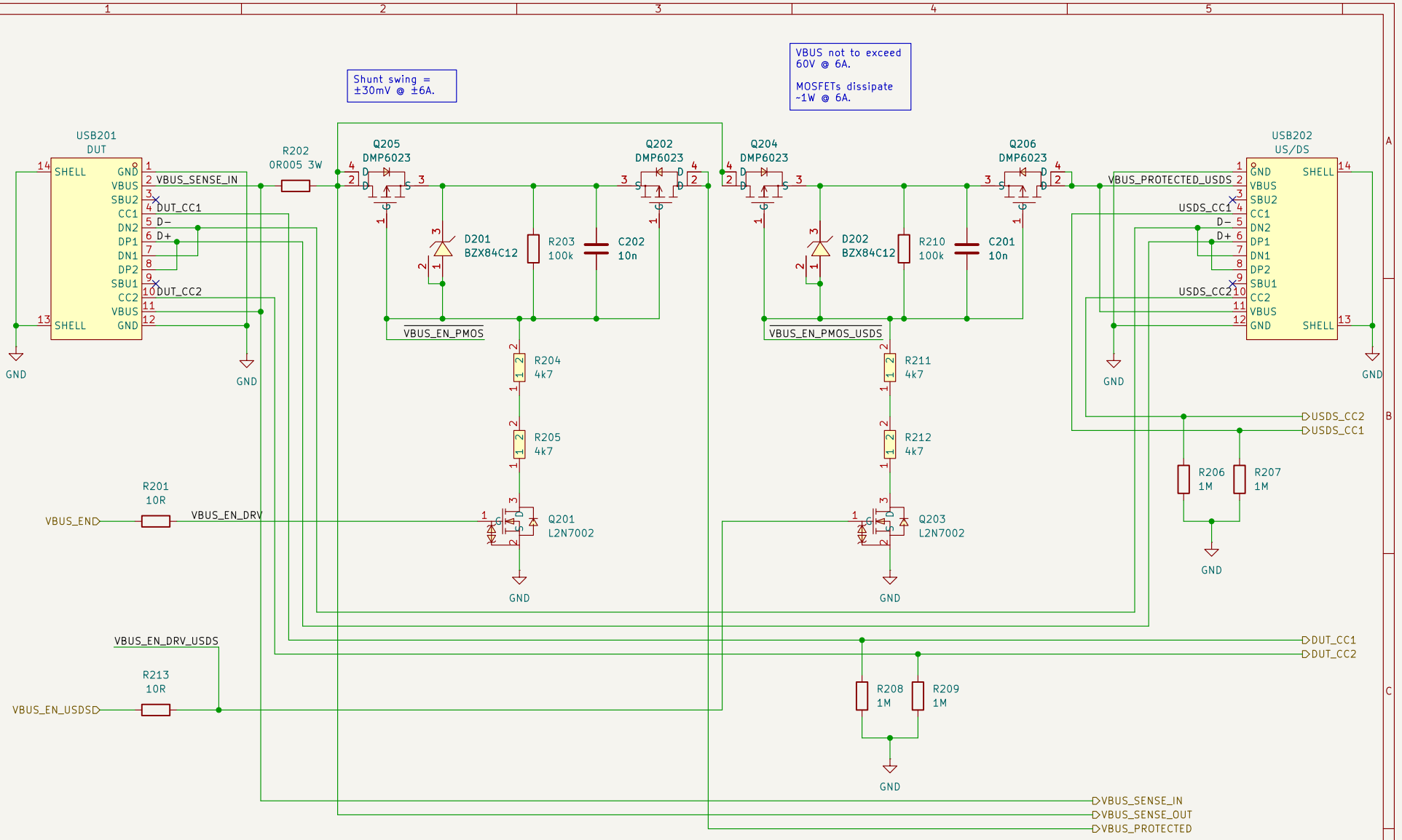

USB Ports
VBUS_SENSE_IN
VBUS_SENSE_OUT
VBUS_PROTECTED
VBUS_EN
VBUS_EN_USDS
DUT_CC1
DUT_CC2
USDS_CC1
USDS_CC2
File: USB Ports.kicad_sch
Analog Monitor
VBUS_SENSE_IN
VBUS_SENSE_OUT
VBUS_VSENSED
VBUS_ISENSED
CC_VSENSED
DUT_CC1
DUT_CC2
USDS_CC1
USDS_CC2
CC_VSENSE_SEL_0
CC_VSENSE_SEL_1
CC_VSENSE_SEL_2
File: Analog Monitor.kicad_sch
Digital Control
VBUS_VSENSE
VBUS_ISENSE
CC_VSENSE
CC_IN
CC_OUT
CC_OUT_EN
CC_VREF_PWM
CC_VSENSE_SEL_0
CC_VSENSE_SEL_1
CC_VSENSE_SEL_2
DUT_CC_SEL
USDS_CC_SEL
USDS_CC_EN
CC1_ROLE_SEL_0
CC1_ROLE_SEL_1
CC1_ROLE_SEL_2
CC2_ROLE_SEL0
CC2_ROLE_SEL1
CC2_ROLE_SEL2
VBUS_EN
VBUS_EN_USDS
SYNC
CC1_VCONN_EN
CC2_VCONN_EN
File: Digital Control.kicad_sch
Bus Connector and Digital IO
DUT_CC_SEL
DUT_CC1
DUT_CC2
USDS_CC_SEL
USDS_CC_EN
USDS_CC1
USDS_CC2
CC_OUT
CC_OUT_EN
CC_VREF_PWM
CC_UNBUF
CC_IND
File: Bus Connector and Digital IO.kicad_sch
PD Role Resistors
DUT_CC1
DUT_CC2
CC1_ROLE_SEL_0
CC1_ROLE_SEL_1
CC1_ROLE_SEL_2
CC2_ROLE_SEL0
CC2_ROLE_SEL1
CC2_ROLE_SEL2
CC1_VCONN_EN
CC2_VCONN_EN
File: PD Role Resistors.kicad_sch
CC_VSENSE_SEL_[0..2]
CC1_ROLE_SEL_[0..2]
CC2_ROLE_SEL[0..2]
Size VBUS_PROTECTED line to CN101 for 6A.
R101 10R
R102 10R
SYNC_DRV
Q101 DMP6023
VBUS_PROTECTED
D103 BZX84C12
R103 100k
GND
CN101 VBUS+
CN102 VBUS−
D102 IP4292CZ10-TBR.115
D101 IP4292CZ10-TBR.115
RF101 CC
RF102 SYNC
GND
All voltage ranges computed based on nominal source voltage.
All resistors are 1206 .125W, ±1% or better unless otherwise noted.
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Sheet: /
File: Dr.PD.kicad_sch
Title: Dr. PD
Size: A
Date: 2026-08-17
Rev: 2608-A
KiCad E.D.A. 10.0.5
Id: 1/6

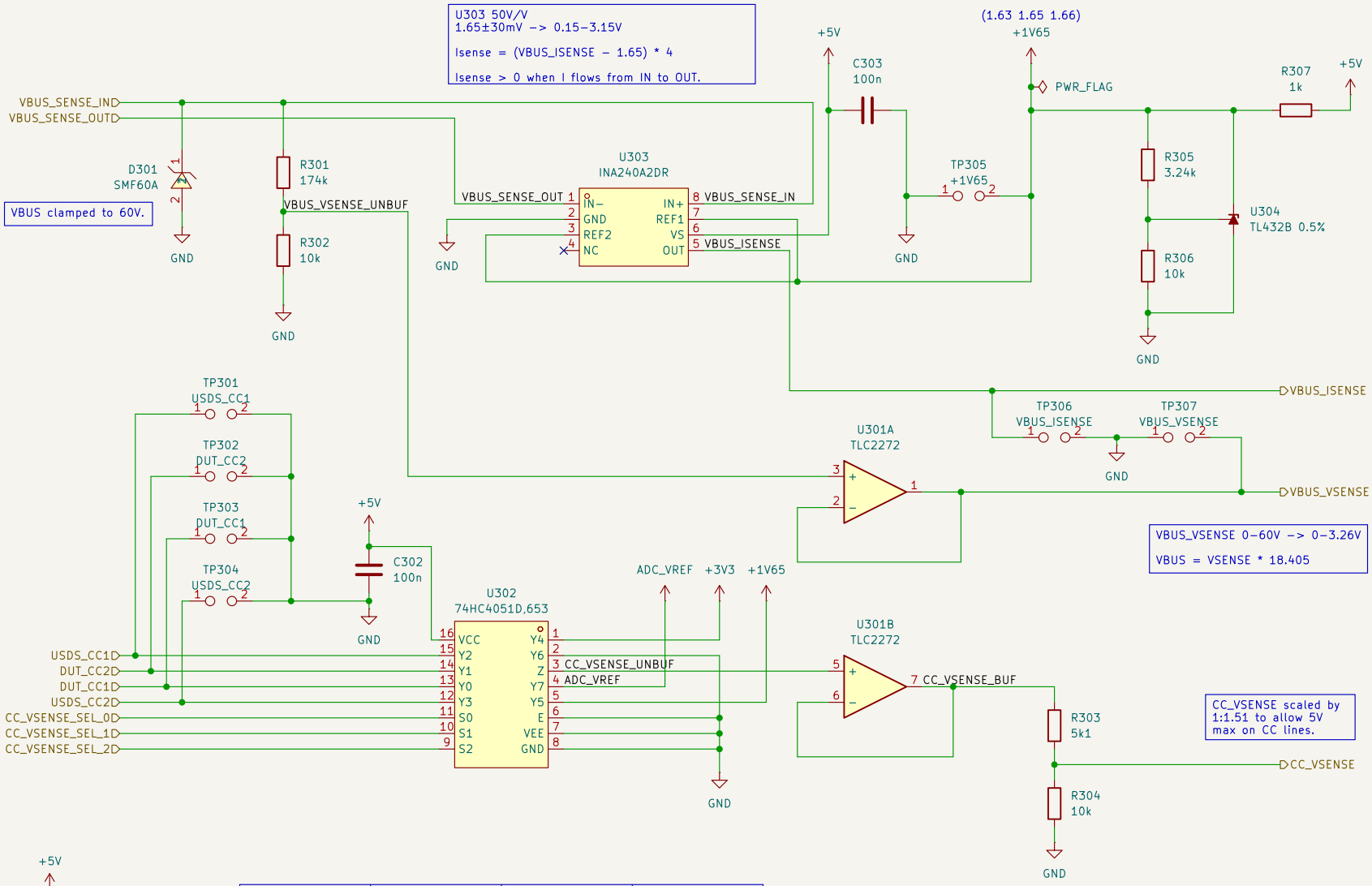


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Sheet: /USB Ports/
File: USB Ports.kicad_sch

Title: USB Ports

Size: A	Date: 2026-08-17	**Rev: 2608-A**
KiCad E.D.A. 10.0.5		Id: 2/6



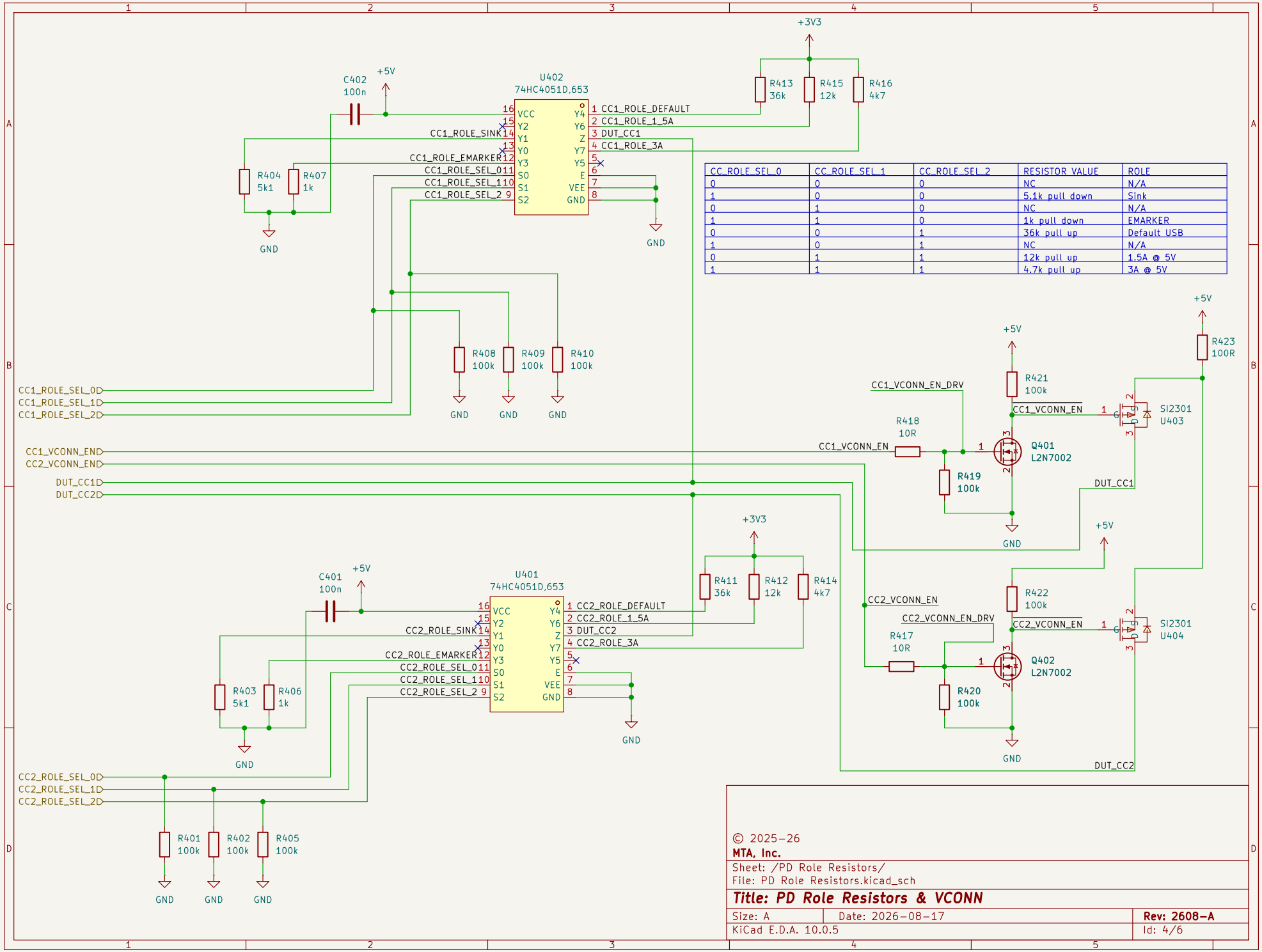
CC_VSENSE_SEL_0	CC_VSENSE_SEL_1	CC_VSENSE_SEL_2	SELECTION
0	0	0	DUT_CC1
1	0	0	DUT_CC2
0	1	0	USDS_CC1
1	1	0	USDS_CC2
0	0	1	+3V3
1	0	1	1.65V VREF
0	1	1	GND
1	1	1	3.3V ADC VREF

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Sheet: /Analog Monitor/
File: Analog Monitor.kicad_sch

Title: Analog Monitor

Size: A	Date: 2026–08–17	Rev: 2608–A
KiCad E.D.A. 10.0.5		Id: 3/6



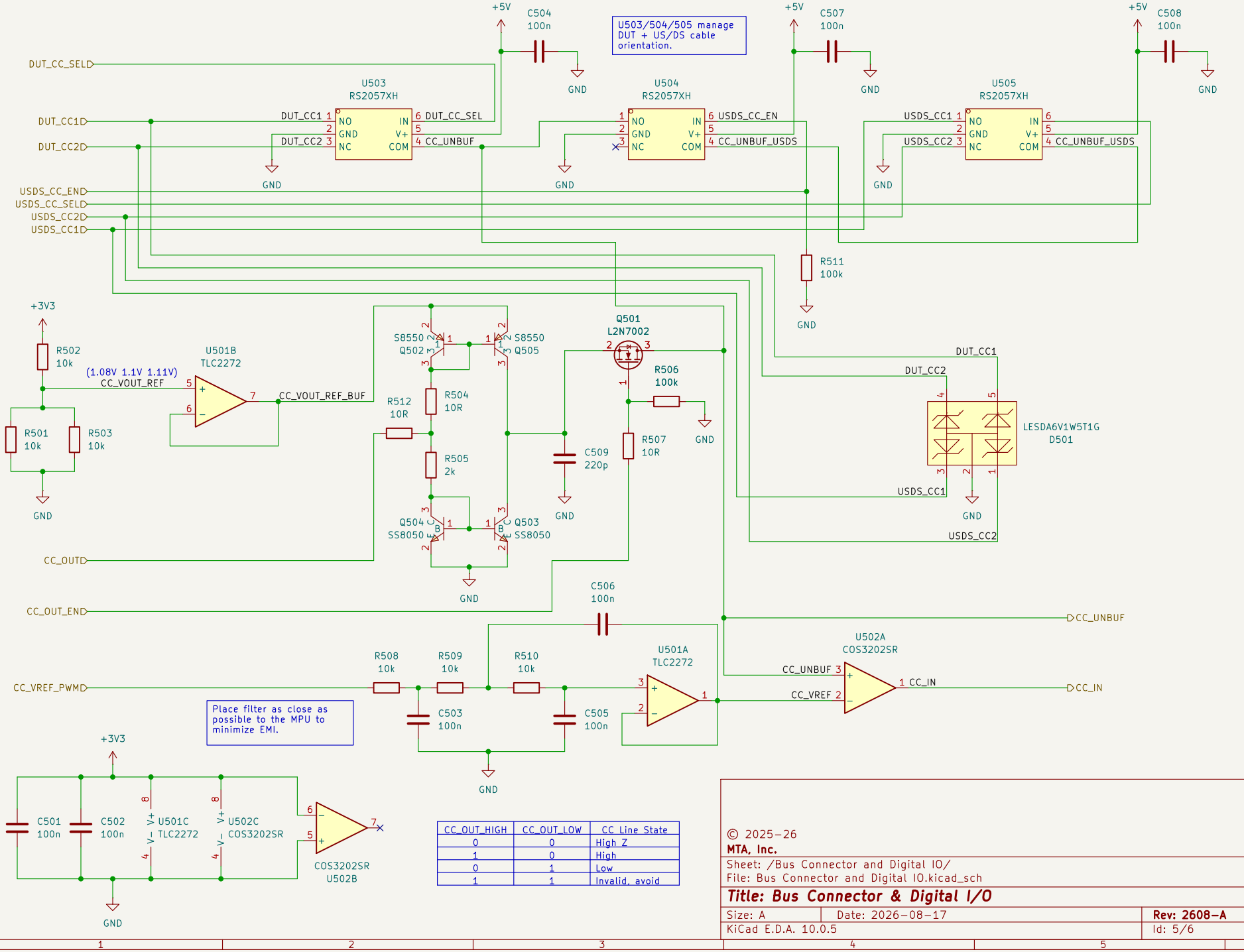
CC_ROLE_SEL_0	CC_ROLE_SEL_1	CC_ROLE_SEL_2	RESISTOR VALUE	ROLE
0	0	0	NC	N/A
1	0	0	5.1k pull down	Sink
0	1	0	NC	N/A
1	1	0	1k pull down	EMARKER
0	0	1	36k pull up	Default USB
1	0	1	NC	N/A
0	1	1	12k pull up	1.5A @ 5V
1	1	1	4.7k pull up	3A @ 5V

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Sheet: /PD Role Resistors/
File: PD Role Resistors.kicad_sch

Title: PD Role Resistors & VCONN

Size: A	Date: 2026-08-17	**Rev: 2608-A**
KiCad E.D.A. 10.0.5		Id: 4/6



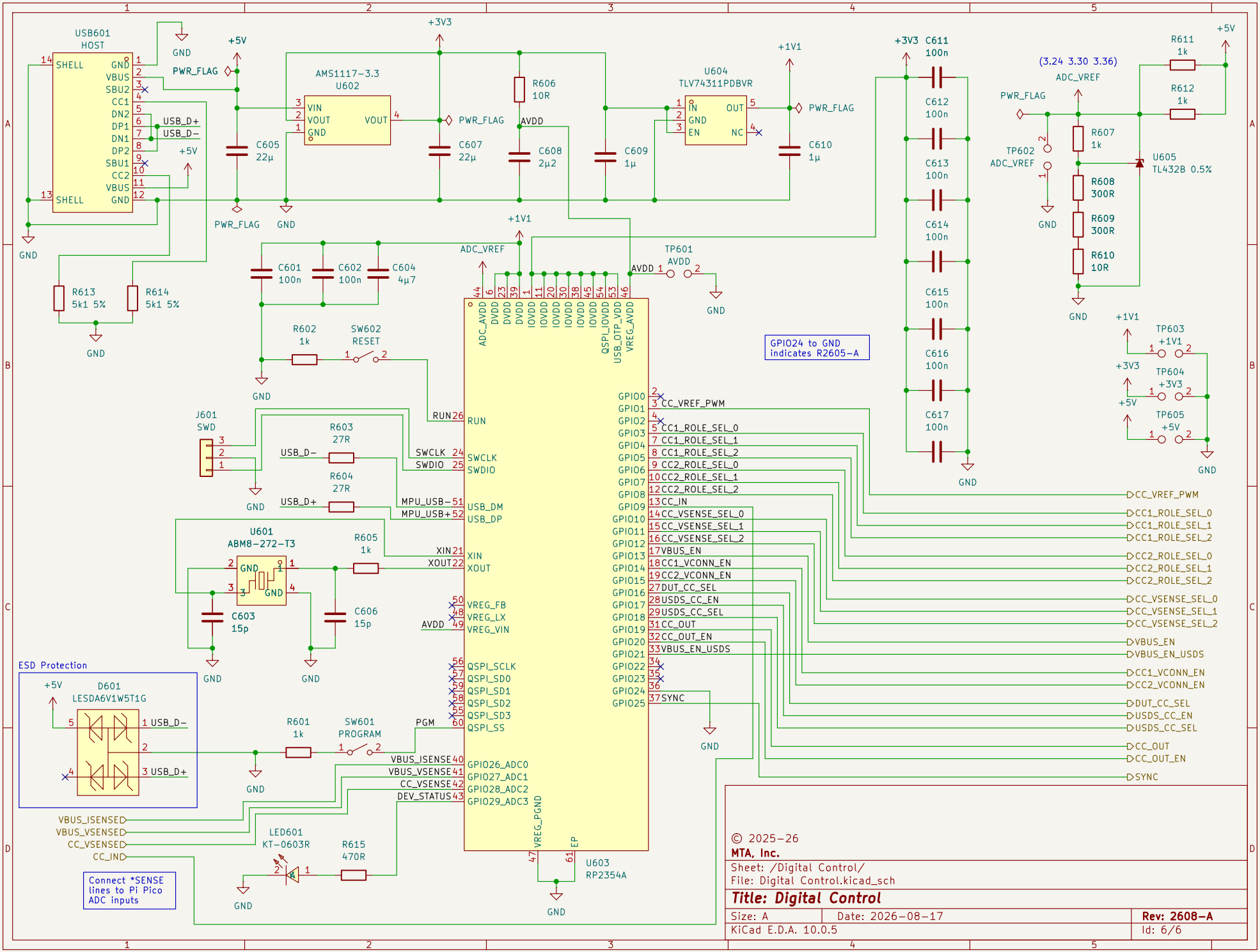
CC_OUT_HIGH	CC_OUT_LOW	CC Line State
0	0	High Z
1	0	High
0	1	Low
1	1	Invalid, avoid

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Sheet: /Bus Connector and Digital IO/
File: Bus Connector and Digital IO.kicad_sch

Title: Bus Connector & Digital I/O

Size: A	Date: 2026-08-17	Rev: 2608-A
KiCad E.D.A. 10.0.5		Id: 5/6


USB601
HOST
SHELL
GND
VBUS
SBU2
CC1
DN2
DP1
DN1
DP2
SBU1
CC2
VBUS
SHELL
GND
USB_D+
USB_D-
PWR_FLAG
+5V
GND
C605
22µ
AMS1117-3.3
U602
VIN
VOUT
GND
+3V3
C607
22µ
R606
10R
AVDD
C608
2µ2
C609
1µ
U604
TLV74311PDBVR
IN
OUT
GND
EN
NC
+1V1
C610
1µ
C611
100n
C612
100n
C613
100n
C614
100n
C615
100n
C616
100n
C617
100n
R611
1k
R612
1k
(3.24 3.30 3.36)
ADC_VREF
R607
1k
R608
300R
R609
300R
R610
10R
TP602
ADC_VREF
U605
TL432B 0.5%
R613
5k1 5%
R614
5k1 5%
C601
100n
C602
100n
C604
4µ7
TP601
AVDD
R602
1k
SW602
RESET
J601
SWD
R603
27R
R604
27R
USB_D-
USB_D+
RUN
SWCLK
SWDIO
MPU_USB-
MPU_USB+
USB_DM
USB_DP
U601
ABM8-272-T3
R605
1k
XIN
XOUT
C603
15p
C606
15p
VREG_FB
VREG_LX
VREG_VIN
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SS
PGM
R601
1k
SW601
PROGRAM
VBUS_ISENSE
VBUS_VSENSE
CC_VSENSE
DEV_STATUS
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
VREG_PGND
EP
U603
RP2354A
ADC_AVDD
DVDD
IOVDD
QSPI_IOVDD
USB_OTP_VDD
VREG_AVDD
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
CC_VREF_PWM
CC1_ROLE_SEL_0
CC1_ROLE_SEL_1
CC1_ROLE_SEL_2
CC2_ROLE_SEL_0
CC2_ROLE_SEL_1
CC2_ROLE_SEL_2
CC_IN
CC_VSENSE_SEL_0
CC_VSENSE_SEL_1
CC_VSENSE_SEL_2
VBUS_EN
CC1_VCONN_EN
CC2_VCONN_EN
DUT_CC_SEL
USDS_CC_EN
USDS_CC_SEL
CC_OUT
CC_OUT_EN
VBUS_EN_USDS
SYNC
GPIO24 to GND
indicates R2605-A
TP603
+1V1
TP604
+3V3
TP605
+5V
ESD Protection
D601
LESDA6V1W5T1G
USB_D-
USB_D+
LED601
KT-0603R
R615
470R
VBUS_ISENSED
VBUS_VSENSED
CC_VSENSED
CC_IND
Connect *SENSE
lines to Pi Pico
ADC inputs
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Sheet: /Digital Control/
File: Digital Control.kicad_sch
Title: Digital Control
Size: A
Date: 2026-08-17
Rev: 2608-A
KiCad E.D.A. 10.0.5
Id: 6/6